

REPORT

Recommendation System for Personalized Content Discovery

Abstract

A personalized movie recommendation system was developed using the Netflix Prize Dataset. Item-Based Collaborative Filtering and Singular Value Decomposition (SVD) were implemented and evaluated using RMSE and MAP@10. SVD achieved the best performance (RMSE = 0.98, MAP@10 = 0.7615), demonstrating the effectiveness of matrix factorization for personalized recommendations.

1. Problem Understanding

As content libraries grow, users often struggle to find content that matches their interests. Recommendation systems solve this problem by analyzing user behavior and suggesting relevant items.

This project aimed to build a personalized movie recommendation engine using historical ratings from the Netflix Prize Dataset, a large-scale real-world dataset of user–movie interactions.

The developed recommendation system was required to:

- Learn user preferences from historical ratings.
- Predict ratings for unseen movies.
- Generate personalized movie recommendations.
- Compare multiple recommendation techniques.
- Evaluate recommendation quality using standard metrics.

Two recommendation approaches were explored:

1. Item-Based Collaborative Filtering
2. Singular Value Decomposition (SVD)

The final goal was not only accurate rating prediction but also effective recommendation ranking.

2. Dataset Description

The Netflix Prize Dataset is one of the most widely used benchmark datasets for recommendation system research.

Dataset characteristics include:

- Users: 480,189
- Movies: 17,770
- Ratings: 100,480,507
- Rating Scale: 1–5
- Additional Information: Rating timestamps and movie metadata

Each rating record contains: User ID, Movie ID, Rating, Date of Rating

For experimentation, a random sample of 1 million ratings was extracted. After filtering users with fewer than 20 ratings, the final modeling dataset contained 50,954 ratings from 1,926 active users.

3. Exploratory Data Analysis

EDA was performed to understand user behaviour, movie popularity patterns, rating distributions, and dataset sparsity. Understanding these characteristics is essential for selecting suitable recommendation techniques.

3.1 Rating Distribution

Most ratings were positive, with ratings of 3, 4, and 5 occurring far more frequently than 1 and 2. Rating 4 was the most common, indicating a positive rating bias and generally high user satisfaction.

3.2 User Activity Analysis

User activity was analyzed by counting the number of ratings provided by each user.

Key observations:

- Most users rated only a small number of movies.
- A small number of highly active users contributed a large portion of the ratings.
- The distribution followed a long-tail pattern, which is common in recommendation systems.

Statistics obtained from the sample:

Total Users:	470,758
Average Ratings per User:	51.10
Median Ratings per User:	24
Maximum Ratings by a Single User:	4,467

This imbalance creates challenges for recommendation systems because learning preferences becomes difficult for less active users.

3.3 Movie Popularity Analysis

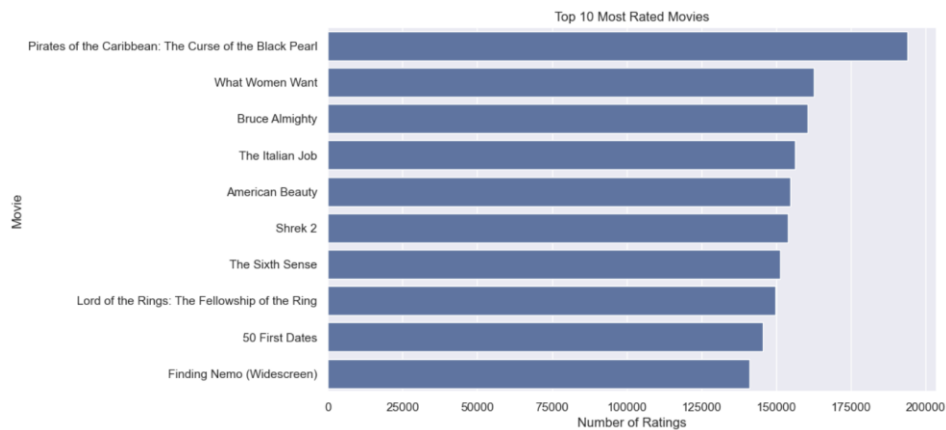
Movie popularity was analyzed by counting the number of ratings received by each movie.

Results showed that a small number of movies received extremely large numbers of ratings while most movies received relatively few ratings.

Movie rating statistics:

- Total Movies: 4,499
- Average Ratings per Movie: 5,346
- Median Ratings per Movie: 552
- Maximum Ratings for a Movie: 193,941

Top 10 most rated movies included:



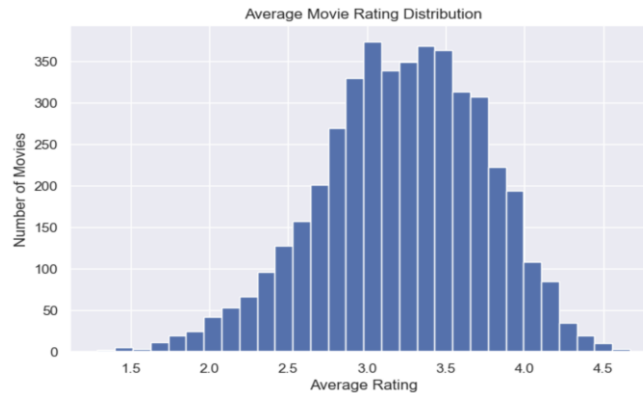
These highly popular movies are likely to appear frequently in collaborative filtering models due to their large interaction histories.

3.4 Average Movie Rating Distribution

The average movie ratings were approximately normally distributed between 2.5 and 4.0.

Most movies achieved average ratings around 3.0–3.5, indicating moderate overall satisfaction among users.

Very few movies achieved extremely high or extremely low average ratings.



3.5 Sparsity Analysis

Sparsity is one of the most important characteristics of recommendation datasets.

The sparsity of the dataset was calculated as:

Users:	470,758
Movies:	4,499
Actual Ratings:	24,053,764
Possible Ratings:	2,117,940,242
Dataset Sparsity:	98.8%

Key EDA Findings

- Users generally provide positive ratings.
- User activity follows a long-tail distribution.
- Movie popularity is highly skewed.
- The dataset exhibits extreme sparsity (98.8%).
- Collaborative filtering methods are suitable because sufficient historical interaction data exists.
- Matrix factorization methods are expected to perform well under sparse conditions.

4. Methodology and Model Design

Two recommendation approaches were implemented and compared.

4.1 Item-Based Collaborative Filtering

Item-Based Collaborative Filtering recommends movies based on similarity between items.

The underlying assumption is that users who liked a particular movie are likely to enjoy movies that are similar to it.

Implementation details:

- Algorithm: K-Nearest Neighbors (KNNBasic)
- Similarity Measure: Cosine Similarity
- Approach: Item-Based
- Rating Scale: 1–5

The recommendation process consists of:

1. Computing similarity between movies.
2. Identifying neighboring movies.
3. Predicting ratings using ratings from similar items.

Advantages:

- Easy to understand and implement.
- Generates explainable recommendations.

Limitations:

- Struggles with sparse datasets.
- Performance decreases when interaction history is limited.

4.2 Singular Value Decomposition (SVD)

Singular Value Decomposition is a matrix factorization technique that learns latent relationships between users and movies.

Instead of relying directly on observed similarities, SVD projects users and movies into a lower-dimensional latent space.

Implementation details:

- Algorithm: SVD
- Number of Factors: 100
- Epochs: 20
- Random State: 42

SVD predicts using hidden factors that represent user preferences and movie characteristics.

Advantages:

- Handles sparse datasets effectively.
- Learns hidden relationships.
- Produces better generalization on unseen data.

Due to these advantages, SVD is widely used in modern recommendation systems.

4.3 Data Preparation

To enable meaningful evaluation:

- A random sample of 1 million ratings was selected.
- Users with fewer than 20 ratings were removed.
- 1,926 active users remained.
- Final filtered dataset contained 50,954 ratings.

A time-based holdout strategy was adopted:

- Training Ratings: 41,324
- Test Ratings: 9,630

For each user, the latest five ratings were reserved for testing while the remaining ratings were used for training.

This approach better reflects real-world recommendation scenarios where future user behaviour must be predicted using historical interactions.

5. Evaluation Metrics

The performance of the recommendation models was evaluated using both rating prediction metrics and recommendation ranking metrics.

5.1 Root Mean Squared Error (RMSE)

RMSE measures the difference between actual ratings and predicted ratings.

It is defined as:

$$\text{RMSE} = \sqrt{\left(\frac{1}{N}\right) \sum (\text{actual rating} - \text{predicted rating})^2}$$

A lower RMSE indicates better prediction of accuracy.

RMSE was used to compare the ability of each model to predict unseen user ratings.

5.2 Mean Average Precision at 10 (MAP@10)

While RMSE evaluates rating prediction accuracy, recommendation systems must also rank relevant items near the top of recommendation lists.

MAP@10 measures ranking quality by considering both relevance and position.

For this project:

- Top 10 recommendations were generated for each user.
- A movie was considered relevant if its actual rating was greater than or equal to 3.5.
- Average Precision was computed for each user.
- MAP@10 was obtained by averaging across all users.

Higher MAP@10 values indicate better recommendation quality.

Evaluation Strategy

A time-based train-test split was used.

For every active user:

- Historical ratings were used for training.
- The latest five ratings were reserved for testing.

This evaluation strategy better reflects real-world recommendation scenarios where future user behavior is predicted from past interactions.

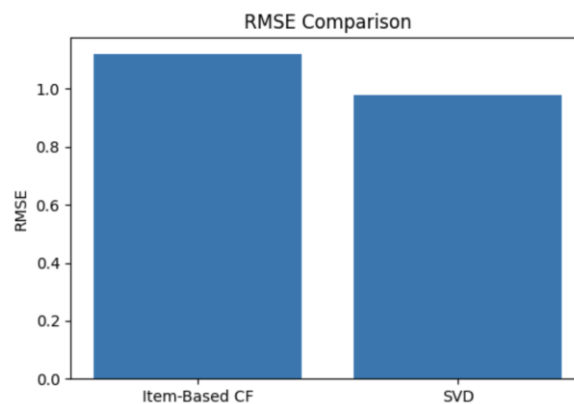
6. Experimental Results

6.1 Rating Prediction Performance

The RMSE scores obtained for both recommendation approaches are shown below.

Model	RMSE
Item-Based Collaborative Filtering	1.12
Singular Value Decomposition (SVD)	0.98

- SVD achieved the lowest RMSE, demonstrating the highest prediction accuracy.
- Its performance comes from learning latent user preferences and movie features.



6.2 Recommendation Ranking Performance

The ranking quality of the final SVD model was evaluated using MAP@10.

MAP@10

0.7615

The obtained MAP@10 score demonstrates that relevant movies were generally ranked near the top of recommendation lists.

Discussion

SVD outperformed Item-Based Collaborative Filtering on the sparse Netflix dataset. By learning latent user–movie relationships, it generated more accurate predictions and was selected as the final recommendation model.

7. Recommendation Examples

To demonstrate recommendation quality, personalized recommendations were generated for a randomly selected user using the trained SVD model.

User Preference Analysis

The user had previously rated the following movies highly:

	title	rating
14	The Hand that Rocks the Cradle	5
19	A Fistful of Dollars	5
4	Urban Cowboy	5
7	The Deal	4
13	From the Earth to the Moon: The Signature Edition	4
11	The Bourne Supremacy	4
23	Shoot to Kill	4
8	Sweet November	4
26	The Incredible Hulk: The TV Series Ultimate Co...	4
27	The Family Man	4

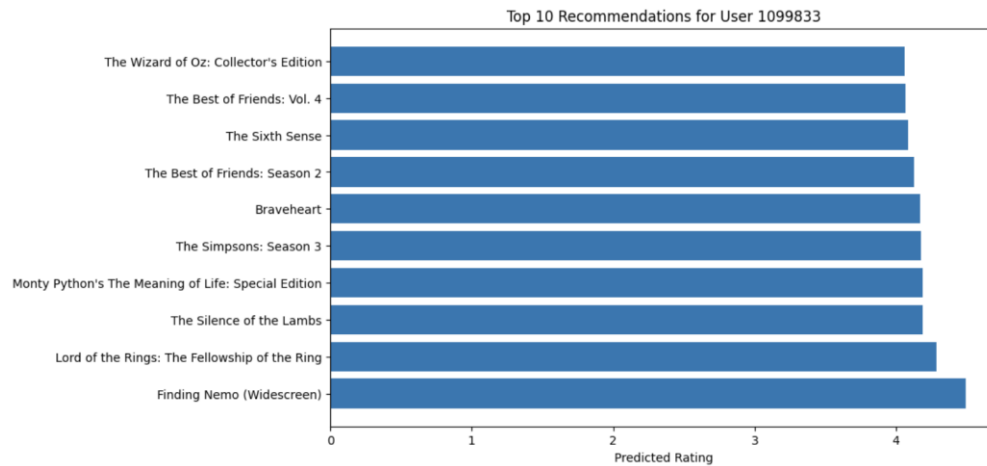
These ratings suggest a preference for action, thriller, drama, and classic films.

Top Recommended Movies

The recommendation system generated the following recommendations:

	movie_id	predicted_rating	year	title
0	3962	4.494001	2003.0	Finding Nemo (Widescreen)
1	2452	4.286506	2001.0	Lord of the Rings: The Fellowship of the Ring
2	2862	4.190422	1991.0	The Silence of the Lambs
3	494	4.187475	1983.0	Monty Python's The Meaning of Life: Special Ed...
4	2172	4.178171	1991.0	The Simpsons: Season 3
5	2782	4.171390	1995.0	Braveheart
6	3078	4.126121	1994.0	The Best of Friends: Season 2
7	4306	4.087171	1999.0	The Sixth Sense
8	1256	4.068854	1994.0	The Best of Friends: Vol. 4
9	3605	4.059069	1939.0	The Wizard of Oz: Collector's Edition

Top 10 personalized movie recommendations generated using the SVD model.



Success Case

Recommended titles such as *The Silence of the Lambs*, *Braveheart*, and *The Lord of the Rings* align with the user's interests, demonstrating the model's ability to capture latent preferences and generate relevant recommendations.

Failure Case

Some recommendations, such as *Friends* and *The Simpsons*, may not fully align with the user's movie preferences. This highlights a limitation of collaborative filtering, which relies on interaction patterns rather than content characteristics.

8. Key Insights

Key findings include the highly sparse nature of the Netflix dataset (98.8% sparsity), long-tail user activity, and concentrated movie popularity. Among the evaluated models, SVD achieved the best performance (RMSE = 0.98, MAP@10 = 0.7615) and successfully generated personalized recommendations.

9. Future Improvements

Several improvements can further enhance recommendation quality.

- Hybrid recommendation systems combining collaborative and content-based filtering.
- Deep learning recommendation models such as Neural Collaborative Filtering.
- Explainable recommendation techniques.
- Improved cold-start handling for new users and new movies.
- Incorporation of additional metadata such as genres, cast information, and user demographics.
- Deployment as a web application or recommendation API.

Conclusion

- Developed a personalized movie recommendation system using the Netflix Prize Dataset.
- Implemented and compared Item-Based Collaborative Filtering and SVD models.
- SVD achieved better prediction accuracy and recommendation quality.
- Demonstrated the effectiveness of latent factor models for sparse datasets.
- Showed how recommendation systems improve content discovery and user experience.

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